



SUNFALSER
OPTOELECTRONIC SOLUTIONS

SFA200AH

Laser Rangefinder Module



Product Manual

Version 202606V1

Disclaimer

This product manual is intended to provide reference specifications and performance parameters of the product. If certain specifications are not included in this document, please contact SunFlaser Technology (Chengdu) Co., Ltd. directly for further confirmation. All contents of this manual and the specifications of the ranging module are subject to change without prior notice.

Without the prior written consent of SunFlaser Technology (Chengdu) Co., Ltd., this document may not be reproduced, stored in any retrieval system, transmitted, or used for any unauthorized purpose in any form or by any means. This document may not be disclosed or distributed to any third party without the explicit authorization of SunFlaser Technology (Chengdu) Co., Ltd.

Due to variations in actual product performance under different operating conditions, the specifications and parameters provided in this document are for reference purposes only and do not constitute any form of warranty. Any express or implied warranties, including but not limited to warranties of merchantability or fitness for a particular purpose, shall be subject to the actual performance of the specific product.

SunFlaser Technology (Chengdu) Co., Ltd. reserves the right to revise or update this document at any time without prior notice. This document may contain technical inaccuracies or typographical errors. Feedback and suggestions are welcome, and verified issues will be corrected in subsequent revisions of the manual.

At the time of publication, all descriptions, specifications, designs, and related procedures contained herein are considered valid. SunFlaser Technology (Chengdu) Co., Ltd. reserves the right to modify the above information at any time without prior notice. If you have any questions regarding the use of this manual, please contact us promptly.

Safety Instructions

For the safe use of the ranging module, please carefully follow the safety instructions provided in this manual. Do not attempt to disassemble, modify, or repair any part of the ranging module. Do not tamper with or attempt to alter the performance of the ranging module.

Intended Use

- Take precautions against electrostatic discharge (ESD). Do not touch any electronic components or parts of the ranging module without proper protective measures.
- Keep this manual readily accessible to all personnel operating the ranging module.
- Operate the ranging module only within the specified voltage and power range.
- Operate the ranging module in a clean, dry, and ESD-protected environment.
- Do not touch the optical glass lens with your fingers.
- Avoid exposing the ranging module to environments with drastic temperature fluctuations.

Improper Use

- Using the ranging module with the host system before verifying that all functions are operating properly;
- Operating the ranging module without first reading and understanding this manual and all safety instructions;
- Disassembling, modifying, or altering the ranging module without authorization from SunFlaser Technology (Chengdu) Co., Ltd.;
- Sending operating commands other than those specified in this manual;
- Using or testing the ranging module in non-standard, abnormal, or unsuitable environments;

- Operating the ranging module in a manner inconsistent with the instructions provided in this manual;
- Continuing to operate the ranging module after improper handling or misuse;
- Using a ranging module with obvious faults, defects, or damage;
- Performing test operations within the blind zone of the ranging module.

Improper use may result in the following risks:

- Eye injury
- Incorrect results
- Measurement errors
- Property damage
- Equipment malfunction

Operator Responsibilities

- Be familiar with the basic knowledge required for the correct operation of the ranging module;
- Understand and comply with the operation and maintenance requirements of the ranging module;
- Prevent damage to the ranging module during storage and transportation;
- Keep the ranging module clean and well maintained;
- Ensure that the ranging module is stored and operated in a safe environment;
- Do not expose the ranging module to strong mechanical shock or impact;
- Do not expose the ranging module to environments with drastic temperature fluctuations.

Avoid Measurement Errors

- Pay attention to all factors that may affect the performance of the ranging module;

- After the Laser Rangefinder Module has been subjected to improper handling (such as vibration, impact, or dropping), or before performing critical measurement tasks, it is recommended to verify the performance specifications of the ranging module.

Table of Contents

01 Product Overview	6
02 Key Features	8
03 Key Specifications	9
04 Interface and Connection	11
05 Communication Protocol and Commands	14
06 Operating Instructions	15
07 Precautions for Use	17
08 Optical Window Requirements	18
9 Safety Precautions	20
10 Maintenance Instructions	20
Packing List	20
Module Cleaning	21
Inspection and Maintenance	21
Troubleshooting	22
Packaging and Transportation Requirements	28
11 Technical Support and Service	30
12 Contact Information	30
Appendix 1	31

01

Product Overview

The SFA200AH Miniature Laser Rangefinder (hereinafter referred to as the laser rangefinder) is a precision optoelectronic product that emits laser pulses toward the target and calculates distance information based on the laser time-of-flight principle.

The laser rangefinder communicates through a TTL or RS422 (selectable) communication interface, featuring excellent performance and simple operation.

The product mainly consists of the following components:

- a) Laser driver circuit assembly;
- b) Detector driver circuit assembly;
- c) Control and information processing circuit assembly;
- d) Transmitting optical assembly;
- e) Receiving optical assembly.

The overall dimensions are shown in Figure 2. The weight is ≤ 50 g, and its appearance is shown below.

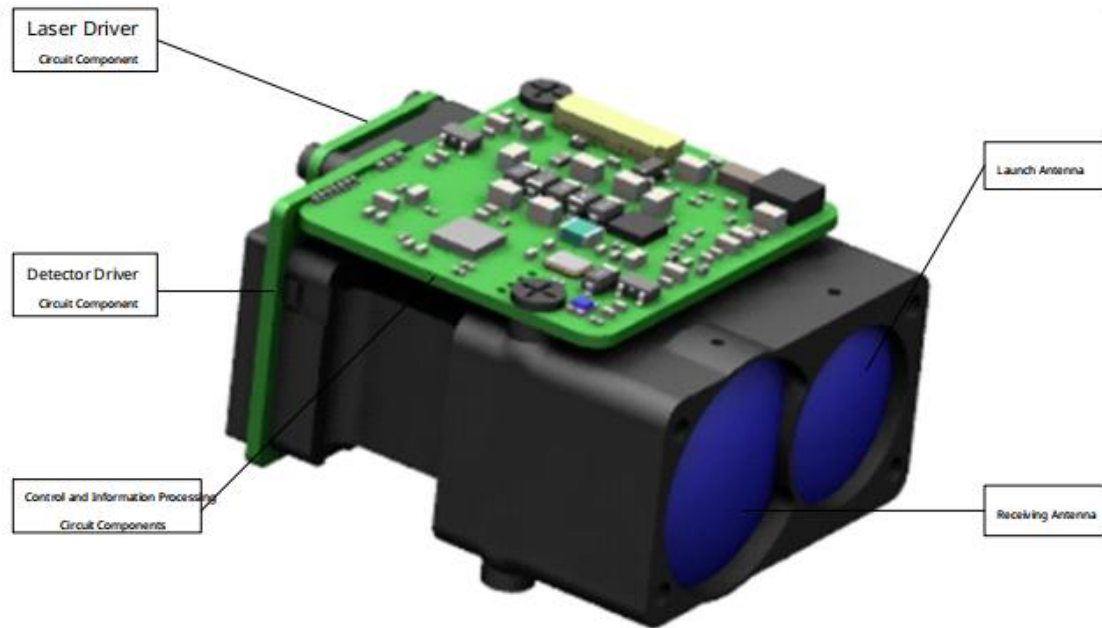


Figure 1 Appearance Schematic

02

Key Features

> **Precision Optoelectronic Ranging**

Utilizing the laser time-of-flight (ToF) principle, it enables high-precision distance measurement.

> **Dual Communication Interfaces Available**

Supports both TTL (3.3 V) and RS422 communication interfaces, allowing users to select the appropriate interface according to system requirements.

> **Wide Voltage Power Supply**

Supports a wide supply voltage range of 9–36 V (24 V typical), compatible with various power supply platforms.

> **Low Power Consumption**

With operating power consumption of less than 2 W, it is suitable for long-duration operation scenarios.

> **High-Frequency Data Acquisition Capability**

The acquisition frequency is adjustable from 1 to 20 kHz, meeting the requirements of high-speed dynamic measurement applications.

> **Flexible Ranging Modes**

Supports single and continuous ranging modes, with real-time adjustment of ranging frequency and baud rate through commands.

> **Compact and Lightweight Design**

Weighing ≤ 50 g, the compact design enables easy integration into platforms such as UAVs, pods, and handheld devices.

03

Key Specifications

No.	Parameter	Performance Specification
1	Wavelength	905 nm $\pm$ 10 nm @ 25°C
2	Acquisition Frequency	Adjustable from 1 to 20 kHz (The output frequency is returned according to the target reflectivity. When the reflectivity is low, the output frequency is lower than the acquisition frequency.)
3	Output Frequency	$\geq$ 100 Hz
4	Maximum Ranging Distance	$\geq$ 200 m (Target reflectivity $\geq$ 50%, visibility $\geq$ 5 km, relative humidity $\leq$ 60%) $\geq$ 120 m (Target reflectivity $\geq$ 10%, visibility $\geq$ 5 km, relative humidity $\leq$ 60%)
5	Minimum Ranging Distance	$\leq$ 0.05 m
6	Ranging Accuracy	$\leq \pm 0.05$ m ($\leq$ 10 m), $\leq \pm 0.1$ m ($>$ 10 m)
7	Operating Temperature	-40°C to +65°C
8	Storage Temperature	-50°C to +85°C
9	Measurement Success Rate	$>$ 98%
10	Supply Voltage	9–36 V (24 V typical)
11	Operating Power Consumption	$<$ 2 W
12	Communication Interface	3.3 V TTL / RS422 (Selectable)
13	Weight	$\leq$ 50 g

> Ranging Capability Description

Under conditions with visibility ≥ 5 km and relative humidity $\leq 60\%$, the laser rangefinder can achieve a ranging distance of over 200 m with a measurement success rate exceeding 98%, which is also a key indicator for evaluating the performance qualification of the laser rangefinder.

The following figure shows the theoretical relationship between the ranging distance of the laser rangefinder and visibility.

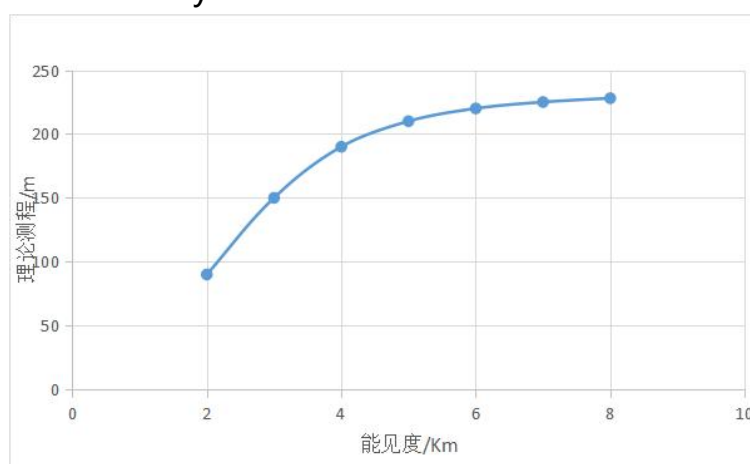


Figure 1 Relationship Between Laser Rangefinder Ranging Distance and Visibilit

Factors such as target reflectivity and measurement angle will affect the ranging capability. Generally, higher target reflectivity results in better ranging performance; the more perpendicular the laser beam is incident on the target surface, the better the ranging performance will be. In practical applications, the laser beam should be directed as perpendicular as possible to the target surface. For medium- to high-reflectivity targets, such as traffic signs, the laser rangefinder can achieve better performance.

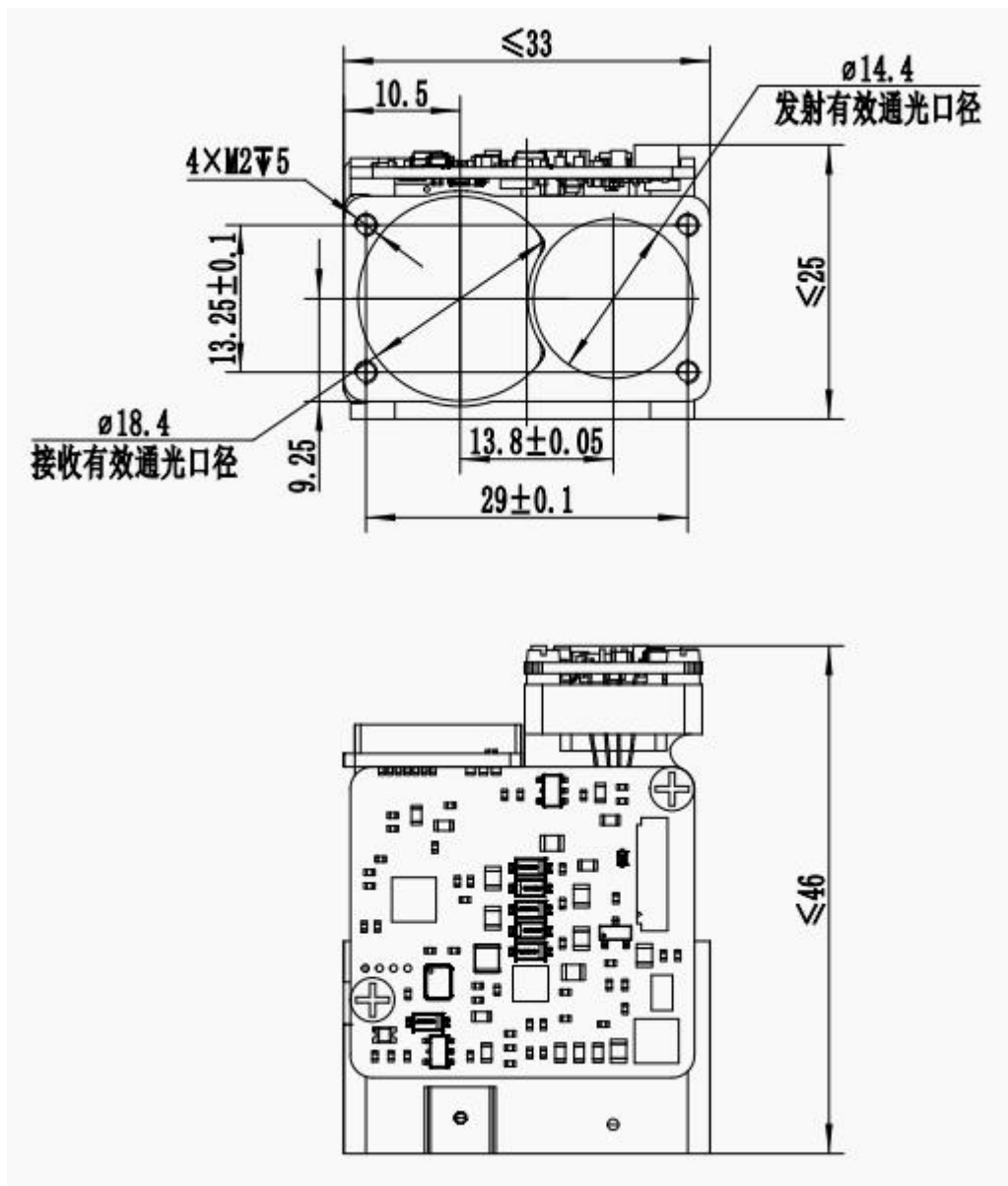
However, for targets that are difficult to produce diffuse reflection, such as water surfaces, the laser rangefinder may not be able to achieve accurate measurements. (For water surface measurement applications, we offer related models that can meet such requirements. Please contact us for more information.)

04

Interface and Connection

Mechanical and Optical Interface:

The mechanical and optical interface dimensions of the laser rangefinder are shown in the figure below. Unit: mm.

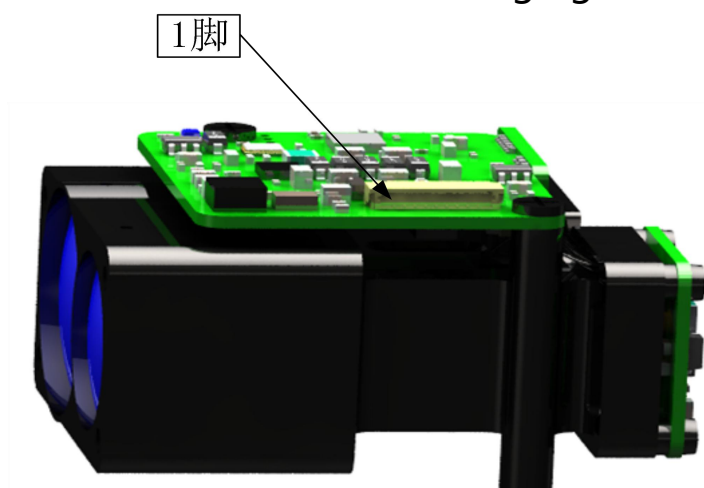


Electrical Interface:

The electrical interface requirements are as follows:

- Supply Voltage: 9–36 V (24 V typical);
- Operating Power Consumption: <2 W;
- Communication Interface: 3.3 V TTL / RS422 (selectable);

The host computer connects to the laser rangefinder through a 10-pin connector for interconnection testing. The pin definitions of the power supply and communication ports on the laser rangefinder side are shown in the table below. The position of Pin 1 on the connector is shown in the following figure.



Pin No.	Label	Electrical Definition	Remarks
P-1	VIN+	Positive power input	Power supply for the laser rangefinder
P-2	VIN+		
P-3	COM	Negative power input	
P-4	COM		
P-5	RS422_TXD+	Signal output port	From the laser rangefinder to the host computer
P-6	RS422_TXD-		
P-7	RS422_RXD-	Signal input port	From the host computer to the

P-8	RS422_RXD+		laser rangefinder
P-9	TTL_RXD	Signal input port	From the host computer to the laser rangefinder
P-10	TTL_TXD	Signal output port	From the laser rangefinder to the host computer

05

Communication Protocol and Commands

For details of the following contents, please refer to Appendix 1:

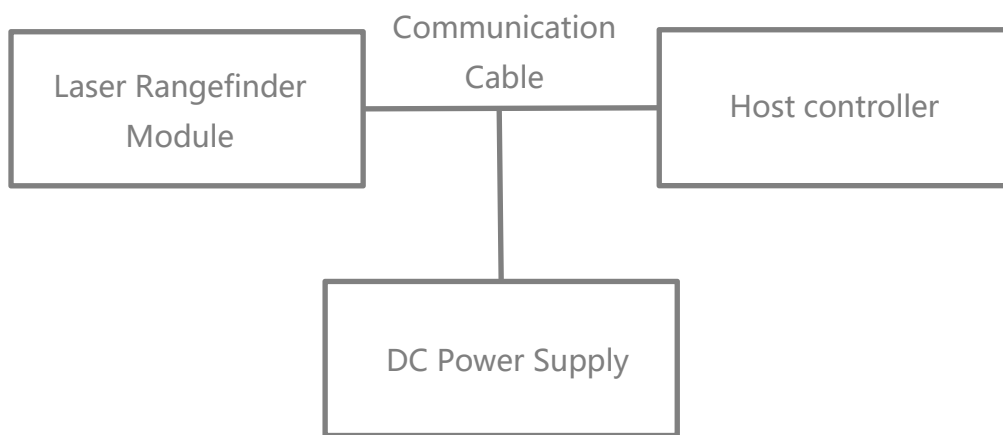
- Communication Interface
- Control Commands Sent to the Laser Rangefinder
- Data Returned by the Laser Rangefinders

06

Operating Instructions

Power-On Procedure

- Before powering on, connect the Laser Rangefinder Module, communication cable, DC power supply, and Host Controller according to the electrical interface requirements and as shown in the figure below.



- The module powers on automatically after the power supply is connected.

Power-Off Procedure

- Before powering off, ensure that all operating processes and tasks have been completed and that the program has been exited properly.
- After confirmation, disconnect the power supply to power off the module normally.

Ranging Mode Operation

- Send the “Start Continuous Measuremen” command (55 03 AA) to the laser rangefinder. The rangefinder will perform continuous ranging and return distance information.
- Send the “Stop Continuous Measuremen” command (55 00 AA) to stop the ranging operation.
- Send the “Set Measurement Frequency” command (55 CF XX XX AA) to adjust the ranging frequency (range: 20 Hz–20 kHz, hexadecimal low byte transmitted first).
- Send the “Set Baud Rate” command (55 F6 XX XX AA) to modify the communication baud rate.
- For detailed command formats and returned data, please refer to Appendix 1.

07

Precautions for Use

- This laser rangefinder is a precision optical product. Improper operation may result in laser hazards. Do not open or adjust any part of the laser rangefinder, and do not attempt to repair or modify its performance by yourself.
- Pay attention to electrostatic protection: electronic components are sensitive to electrostatic discharge. Do not touch any electronic components without proper protective measures.
- The device must only be powered on and operated within the specified voltage range (9–36 V) and power requirements.
- Do not touch the optical lenses with fingers or hard objects (to prevent contamination from grease or scratches on the lens surface).
- Do not measure highly reflective targets at extremely close distances (to prevent damage to the detector's core components).
- Do not store the device under non-specified conditions (such as highly polluted environments or temperatures beyond the specified storage range).
- Do not expose the laser rangefinder to severe mechanical impacts (such as vibration, shock, or drops).
- Avoid using the device under direct sunlight or in environments with rapid temperature changes.
- It is recommended to operate the device in environments with humidity below 80% and keep the operating environment clean.
- Do not attempt to disassemble or modify any part of the laser rangefinder, and do not send any control commands other than those specified in this manual.

08

Optical Window Requirements

> **Material Selection**

The optical window material is recommended to use optical glass H-K9L. H-K9L is one of the most commonly used colorless optical glasses, suitable for laser wavelengths from 300 nm to 2100 nm. It offers excellent physical properties and high cost-effectiveness.

> **Processing Requirements**

- a) The wedge angle tolerance of the optical window should be minimized. A wedge angle tolerance of $\leq 3'$ is recommended (tolerance grade $\leq$ Grade 7).
- b) The optical surfaces of the window should be as smooth as possible. An arithmetic average surface roughness (Ra) of 0.012 is recommended.

> **Coating Recommendations**

For 905 nm laser rangefinder optical windows, an anti-reflection coating within the range of 855 nm–955 nm is recommended, with a transmittance of $\geq 99.5\%$. Depending on the specific application environment, additional protective coatings such as hydrophobic coatings or hard coatings may be applied to the outer surface of the optical window. Other requirements shall comply with GJB 2485A-2019, with a transmittance of $\geq 97\%$.

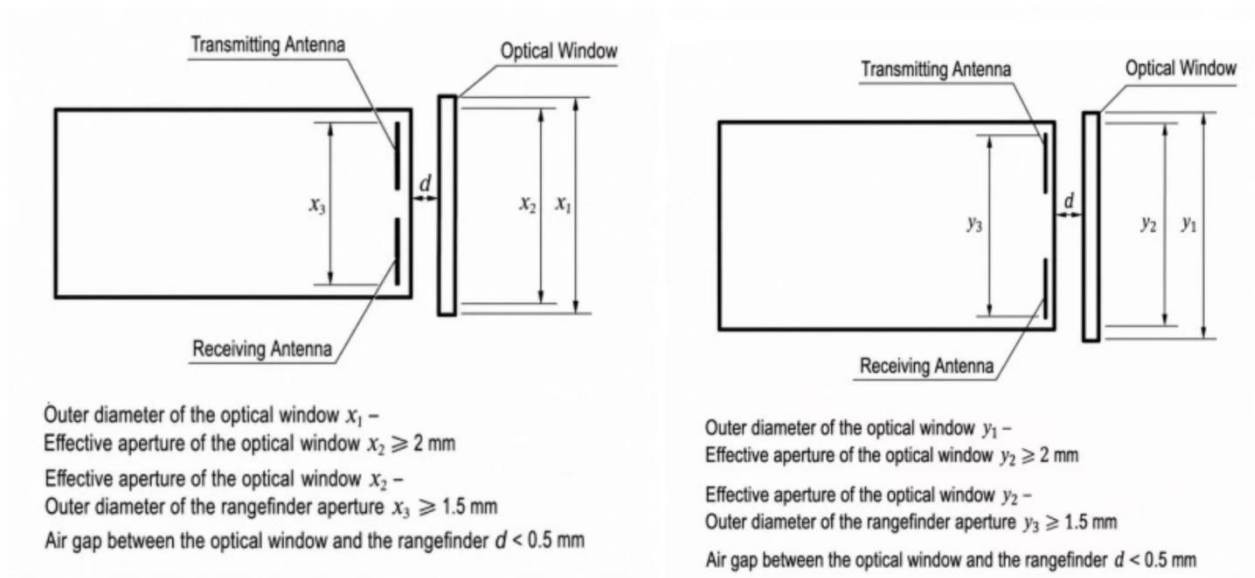
> **Optical Window Design and Usage Recommendations**

The effective aperture of the optical window depends on the specific product. The dimensions should ensure that: optical window outer diameter – effective aperture ≥ 2 mm, and effective aperture – rangefinder optical aperture outer diameter ≥ 1.5 mm. The schematic diagram is shown below.

Since the optical window absorbs a certain amount of laser energy, it is recommended that the window thickness be controlled within 2–4 mm according to the overall dimensions.

Due to the high transmittance of the optical window, it is recommended that the laser emission optical axis be parallel to the optical window normal axis, and that the air gap between the optical window and the rangefinder be less than 0.5 mm. The positional relationship between the optical window and the two lens barrels is shown in the figure below.

During use, the optical window must be kept clean and transparent, with no stains, fingerprints, dust, or other contaminants on the optical surface.



9

Safety Precautions

- To ensure the safety of both operators and test subjects, the following safety measures have been incorporated into the design of the SFA200AH Laser Rangefinder Module;
- Safety design and analysis are conducted in accordance with GJB 900A-2012, General Requirements for Equipment Safety;
- Non-flammable materials are used, and all mechanical and electrical interfaces are securely connected;
- Appropriate design measures are implemented to prevent moisture accumulation that could result in short circuits;
- The module operates below the safety voltage limit for the human body.

10

Maintenance Instructions

Packing List

The standard items supplied with the product are listed in the table below. Optional accessories may not be included in this list and shall be subject to the actual product delivered.

No.	Accessory Name	Quantity	Remarks
1	Anti-static Self-sealing Bag	1	
2	Desiccant	1	
3	Product Certificate of Conformity	1	
4	Communication Connector	1	
5	Packaging Box	1	

Module Cleaning

Cleaning of Optical Components

- Dust particles should be removed using an air blower;
- Fingerprints should first be cleaned with absorbent cotton lightly moistened with a small amount of alcohol-ether mixture, and then wiped with a clean lens cloth.

Cleaning of Mechanical and Electronic Components

- With the power disconnected, gently wipe mechanical and electronic components with alcohol and allow them to dry naturally before use;
- Keep the ranging module, connectors, and cables away from moisture and contaminants whenever possible;
- Ensure that the equipment is completely dry before packaging.

Inspection and Maintenance

Routine Maintenance

When the product is used for the first time, or after replacement of related modules or components, both visual inspection and power-on inspection shall be performed. For products under normal operation, only a power-on inspection is required before use.

- Visual Inspection Procedure
 - Check whether the product appearance is normal;
 - Check whether the cable connections are correct and securely connected.
- Power-On Inspection Procedure
 - Perform the power-on procedure and verify that the standby current is correct;
 - Start the self-test function of the product;
 - After the inspection is completed, perform the power-off procedure.

Periodic Maintenance

Under normal operating conditions, the Laser Rangefinder Module does not require routine maintenance. If the module has been stored in a dust-free environment for more than one year, maintenance should be performed, including general inspection and power-on inspection.

- General Inspection (Performed with Power Off)
 - All markings and serial numbers on the product and test cable connectors shall be correct and clearly legible;
 - All screws on the panel shall be securely fastened;
 - The optical glass of the product shall be free from visible defects that may affect normal operation, such as spots, pitting, water stains, fungus, fingerprints, dust particles, or cracks.
- Comprehensive Power-On Inspection and Maintenance
 - Connect the product power supply in sequence;
 - Perform the power-on procedure and verify that the standby current is correct;
 - Start the product self-test and complete the self-test procedure;
 - Perform the power-off procedure after completion of the inspection.

Troubleshooting

The following list only includes possible fault phenomena not caused by the product itself. If any other technical problems occur during operation, please contact our company for technical support.

Problem Description	Possible Causes of Failure	Troubleshooting Method	Evaluation Criteria	Solution / Corrective Action
No communication or abnormal communication of the rangefinder module	Power supply issue	Check whether the power supply has hardware or software wake-up enabled. Verify that the voltage setting is correct and ensure the protection current is properly configured. If the voltage is correct, try replacing the power supply and test again.	If the module operates normally after changing the voltage or power supply, the issue can be determined to be caused by an incorrect power supply.	Set the correct voltage and current, or replace the power supply.
	Connection cable issue	Check whether the wiring is secure, or replace the communication cable for testing (TTL communication cables should not be excessively long).	If communication is restored after rewiring or replacing the communication cable, the issue can be determined to be caused by a connection cable problem.	Rewire the connections or replace the communication cable.
		Check whether the wiring matches the interface pin definition.	If the module operates normally after confirming that the wiring matches the pin definition, the issue can be determined to be caused by incorrect wiring.	Reconnect the wiring according to the correct pin definition.
	Incorrect baud rate setting	1. Check whether the rangefinder status and baud rate settings are consistent with the communication protocol. 2. If the rangefinder supports baud rate configuration, change to the required baud rate and try communication again.	If communication operates normally after aligning the baud rate with the communication protocol, the issue can be determined to be caused by an incorrect baud rate setting.	Set the host computer baud rate to match the communication protocol.

Problem Description	Possible Causes of Failure	Troubleshooting Method	Evaluation Criteria	Solution / Corrective Action
	Incorrect communication command	Check the rangefinder status and verify whether the transmitted command is consistent with the communication protocol, and confirm that the command is in hexadecimal format.	If communication operates normally after confirming that the transmitted command matches the communication protocol, the issue can be determined to be caused by an incorrect communication command.	Send the correct command to the rangefinder module.
	Main control program issue	If communication fails during integration with the main controller, first check whether standalone communication is functioning normally.	If standalone communication works normally, the issue can be considered likely to be caused by a problem in the main control program.	It is recommended to check whether there are issues in the system-level program (e.g., whether the parity/check bits are correct, and whether the host computer data parsing and display are implemented correctly).
	Other causes (e.g., circuit failure, etc.)	Return to factory for inspection and troubleshooting.	Return to the factory for inspection and diagnosis.	Return to factory for repair.
The rangefinder returns invalid values.	Operational error / improper operation.	Check whether the receiving or transmitting lens is blocked, and ensure that there are no obstructions in the measurement environment (e.g., window glass, wall corners, etc.).	If the rangefinder returns normal readings after removing the obstruction, the issue can be considered likely caused by optical blockage.	Remove the obstruction.
		Whether the ranging axis is aligned with the optical sighting axis.	If the rangefinder returns normal measurements after realignment, the issue can be determined to be caused by misalignment between the ranging axis and the optical sighting axis.	Realign the ranging and optical axes and perform measurement again.

Problem Description	Possible Causes of Failure	Troubleshooting Method	Evaluation Criteria	Solution / Corrective Action
	Laser transmitter failure.	Check whether the current reading is normal. Observe whether the current fluctuates during ranging. Send a self-test command to check the returned data. Alternatively, use an infrared test card or a CCD camera to verify whether a laser spot is present.	If the emission self-test is abnormal and no laser spot is observed, the issue can be preliminarily determined to be caused by a laser transmitter failure.	Return to factory for repair.
	Other causes (e.g., circuit failure, detector, etc.)	Return to factory for inspection and troubleshooting.	Return to the factory for inspection and diagnosis.	Return to factory for repair.
False ranging results.	Electrical noise interference.	Block or point the rangefinder receiving lens toward open space (or away from the sun) and test whether false alarms still occur. If no false alarms are observed, the issue can be determined to be caused by optical noise interference. Further determine whether it is due to external ambient light or the system optical window (e.g., mismatch of wavelength band with the rangefinder, excessive tilt angle of the optical window, small aperture causing light blockage, or a contaminated optical window). Power the rangefinder module independently within the system	It is recommended to address power supply noise issues by adding a larger common-mode filter in series, replacing the grounding capacitor with a 10 nF capacitor, and increasing the detection threshold (note that this may reduce system performance).	
	Communication noise interference.	and test whether false alarms still occur. If no false alarms are observed, the issue can be determined to be caused by electrical or communication noise interference. Power the rangefinder module independently within the system and test whether false alarms still occur. If no false alarms are observed, the issue can be determined to be caused by power supply noise interference. Operate the rangefinder module with communication only	It is recommended to address communication noise interference by adding a ferrite core (shielding magnetic ring) to the cable or increasing the threshold level (note that this may reduce system performance).	
	Optical noise interference.	within the system and test whether false alarms still occur. If no false alarms are observed, the issue can be determined to be caused by communication noise interference. If false alarms still occur after the above steps, test the rangefinder module as a standalone unit (removed from the system and powered/communicated independently). If false alarms are still present, the issue can be determined to be caused by the rangefinder module itself or the power adapter.	It is recommended to address optical noise issues (optical window condition and test angle) and increase the detection threshold (note that this may reduce system performance).	

Problem Description	Possible Causes of Failure	Troubleshooting Method	Evaluation Criteria	Solution / Corrective Action
	Other causes (e.g., circuit failure, etc.).	Return to factory for inspection and troubleshooting.	Return to the factory for inspection and diagnosis.	Return to factory for repair.
Insufficient performance capability of the rangefinder module.	Environmental conditions (weather-related factors).	Check whether the weather conditions include rain, snow, fog, or dust/sand, whether atmospheric visibility is too low, whether humidity is too high, or whether there are localized atmospheric turbulence effects, as well as any discrepancies between the environmental conditions and the test site.	If the ranging performance meets the technical specifications when the visibility is above the required threshold, the issue can be determined to be caused by environmental weather conditions.	Test the rangefinder performance under appropriate weather conditions.
	Optical window issue.	Determine, by checking the system configuration, whether the issue is caused by low optical window transmittance, incorrect wavelength band, contamination, small aperture causing light blocking, or excessive tilt angle. Remove the optical window and test the rangefinder module independently to observe whether its ranging performance is degraded. Also inspect whether the lens of the rangefinder is excessively contaminated.	If the ranging performance meets the specification requirements after removing the optical window, the issue can be determined to be caused by low transmittance of the host optical window. If the ranging performance meets the specification requirements after cleaning the rangefinder lens, the issue can be determined to be caused by low transmittance of the host optical window.	Replace the optical window according to the recommended selection and installation guidelines, or switch to a higher-performance rangefinder model. Clean the lens.

Problem Description	Possible Causes of Failure	Troubleshooting Method	Evaluation Criteria	Solution / Corrective Action
	Target size is too small or target reflectivity is too low.	Replace with another target. It is recommended to use targets such as buildings that are perpendicular to the optical axis, and avoid hollow or perforated targets as much as possible.	If the ranging performance meets the specification requirements after replacing with a higher-reflectivity or larger target, the issue can be determined to be caused by inappropriate target selection.	Replace the target, or switch to a higher-performance rangefinder model.
	Operating temperature exceeds the specified range of the rangefinder module.	When ranging performance degrades, send a self-test command to check the current operating temperature of the rangefinder module. After the temperature returns to normal, retest and observe whether the performance issue persists.	If the operating temperature exceeds the specified range when performance degrades, and the performance issue disappears after the temperature returns to normal, the issue can be determined to be caused by the operating environment exceeding the allowable temperature range of the rangefinder module.	Operate the device within the specified operating temperature range.
	Optical axis misalignment.	Check whether high/low temperature tests, thermal shock tests, shock tests, or vibration tests were performed prior to the degradation of ranging performance, or whether any drop or impact event occurred.	Return to the factory for inspection and diagnosis.	Return to factory for repair.
	Other causes (e.g., circuit failure, low receiver sensitivity, etc.)	Return to factory for inspection and troubleshooting.	Return to the factory for inspection and diagnosis.	Return to factory for repair.

Problem Description	Possible Causes of Failure	Troubleshooting Method	Evaluation Criteria	Solution / Corrective Action
Accuracy out of tolerance.	Large variation in target reflectivity.	Confirm whether the tested target is a high-reflectivity target, and determine whether the issue is large data fluctuation or accuracy out of tolerance.	If the tested target is highly reflective and the ranging accuracy returns to normal after replacing it with a suitable target, the issue can be determined to be caused by excessive target reflectivity variation.	Replace the target. The use of highly reflective test targets is not recommended.
	Test operation issue.	Check whether the target has been properly aimed at.	If the ranging accuracy returns to normal after correctly aiming at the target, the issue can be determined to be caused by improper test operation.	Re-aim at the target.
	Instrument issue.	Verify the accuracy of the calibration instrument (possible calibration error).	If the ranging accuracy returns to normal after replacing with an accurately calibrated instrument, the issue can be determined to be caused by the test instrument.	Replace the calibration instrument.
	Low optical window transmittance	Check whether the ranging value is abnormally high.	If the ranging accuracy returns to normal after removing the optical window, the issue can be determined to be caused by the optical window.	Replace the optical window according to the recommended optical window selection and installation guidelines.

Packaging and Transportation Requirements

> Packaging

After unpacking, if the product needs to be placed back into storage, it shall be repackaged using the original packaging whenever possible.

If the product needs to be returned to the manufacturer, the original packaging is recommended. If alternative packaging is used, it shall not cause any degradation in product performance or physical damage.

> **Transportation**

Repackaged products may be transported by automobile, railway, aircraft, or ship. During transportation, the package shall be securely fixed to the transport vehicle to avoid impact, improper handling, and exposure to rain or snow.

> **Storage**

The product shall be stored in a warehouse with a storage temperature of 0°C to +30 °C, relative humidity not exceeding 80%, free from corrosive substances, strong mechanical vibration and shock, and strong magnetic fields.

11

Technical Support and Service

If a product malfunction occurs and factory return is required for fault analysis, troubleshooting, or repair, SunFlaser Technology (Chengdu) Co., Ltd. provides a one-year warranty and lifetime technical support from the date of product delivery.

During the warranty period, product failures caused by manufacturing or quality issues will be repaired or replaced free of charge by SunFlaser. For product damage caused by improper operation or other user-related factors, repair and replacement costs will be charged according to the actual situation.

This product is a precision optical instrument. Please handle and operate it carefully during use. If you have any additional questions regarding operation or maintenance, please contact our after-sales support personnel at any time.

12

Contact Information

Website: www.sunflaser.com

Tel: +86 186 2827 7862 (Sales Manager: Mr. Cao)

Address: No. 58, Zhenxing West 1st Road, Building 1, Room 201, Jinniu District, Chengdu, Sichuan, China

Appendix 1:

Communication Interface

Baud Rate: Adjustable among 921600 bps / 460800 bps (default) / 256000 bps / 115200 bps / 57600 bps / 38400 bps / 19200 bps.

Since the serial interface has a limited bit transmission capacity per second, different ranging frequencies require different baud rate settings:

- When the acquisition frequency is <200 Hz, any baud rate may be used.
- When the acquisition frequency is >200 Hz, the baud rate must be set to 19200 bps or higher.
- When the acquisition frequency is >400 Hz, the baud rate must be set to 115200 bps or higher.
- When the acquisition frequency is >2500 Hz, the baud rate must be set to 256000 bps or higher.
- When the acquisition frequency is >6000 Hz, the baud rate must be set to 460800 bps or higher.
- When the acquisition frequency is >10500 Hz, the baud rate must be set to 921600 bps.
- Single-byte transmission format: 1 start bit, 8 data bits, No parity, and 1 stop bit. For 8-bit data transmission, the low byte is transmitted first, followed by the high byte.

Communication Protocol

Communication Rate and Format

Format Specification	Baud Rate: 921600 bps / 460800 bps (default) / 256000 bps / 115200 bps / 57600 bps / 38400 bps / 19200 bps Byte Format: 1 start bit, 8 data bits, 1 stop bit, no parity
-----------------------------	--

Control Commands Sent to the Rangefinder

Function	Command (Hex)	Remarks
Start Continuous Measurement	55 03 AA	
Stop Continuous Measurement	55 00 AA	
Set Measurement Frequency	55 CF XX XX AA	XXXX range: [20–20k], entered in hexadecimal, low byte first
Set Baud Rate	55 F6 XX XX AA	XXXX = desired baud rate ÷ 100, entered in hexadecimal, low byte first

Data Returned by the Rangefinder

Power-On Notification and Frequency/Baud Rate Information

Byte No.	Byte Definition (Hex)	Remarks
0	0x5C	Frame Header
1	0xXX	Current Temperature
2-3	0xXX 0xXX	Frequency Information (Low Byte First)
4-5	0xXX 0xXX	Baud Rate Information (Low Byte First; actual baud rate = value × 100)
6	0xHH	One's Complement of the Sum of Bytes 1–5

Distance Information

Byte No.	Byte Definition (Hex)	Remarks
0	0x5C	Frame Header
1-2	0xXX 0xXX	Distance Information (Low Byte First), Unit: cm
3	0xHH	One's Complement of the Sum of Bytes 1–2